IIIT Vadodara-ICD
CODE PLAYGROUND

GROUP MEMBERS

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INTRODUCTION

- Code playground is an interactive coding platform for practicing coding problems, participating in contests, and preparing for technical interviews
- Built for students, coding enthusiasts, and job seekers. programmers
- Accessible, interactive, and beginner-friendly

OBJECTIVES

- Create a clean and user-friendly coding environment
- Provide code execution in multiple languages
- Progress tracking
- Leaderboards
- Contests
- Community interaction

TECH STACK

- **Frontend**: HTML, CSS, JavaScript with EJS templating for dynamic pages
- **Backend**: Built using Node.js and Express.js for efficient server-side logic
- **Database**: MongoDB with Mongoose for structured data storage and retrieval
- **Authentication**: Secure login system using JWT tokens and bcrypt password hashing

KEY FEATURES

- **User Authentication** – Secure login, registration, and personalized profiles
- **Problem Solving** – Practice coding problems sorted by topics and difficulty levels
- **Real-Time Execution** – Write, compile, and run code instantly within the platform
- **Contests & Leaderboards** – Participate in coding competitions and track rankings
- **Community Forums** – Interact, discuss solutions, and learn collaboratively
- **Responsive Design** – Seamlessly accessible on desktops, tablets, and mobile devices

DATA MODELS

- **User**: Username, email, password, solved problems, submissions, contest history
- **Problem**: Title, description, difficulty, test cases, constraints
- **Contest**: Name, description, start/end time, problems, participants, leaderboard
- **Discussion**: Topic, content, author, comments, votes
- **Comment**: Content, author, votes

USER INTERFACE

- **Home Page** – Platform overview and quick access to features
- **Problem Listing** – Filter problems by topic and difficulty
- **Code Editor** – Syntax highlighting and real-time output
- **Contests** – Participate and manage live coding competitions
- **User Profiles** – View stats, submissions, and solved problems
- **Discussion Forums** – Engage with the community and share solution

HOME PAGE

- Clean, welcoming interface highlighting core platform features
- Easy and intuitive navigation to key sections:
- Explore – Overview and latest updates
- Problems – Browse coding questions by category
- Contests – Join live and upcoming competitions
- Discuss – Community discussions and Q&A
- Profile – Track user progress and submissions
- About Us – Learn more about the platform and team

PROBLEM SOLVING WORKFLOW

- **Browse Problems** by difficulty level – Easy, Medium, Hard
- **Select a Problem** to read description, constraints, and sample cases
- **Write Code** in your preferred programming language
- **Test with Custom Inputs** for quick debugging
- **Submit Solution** to evaluate against hidden test cases
- **View Results** including status, runtime, and accuracy

CONTEST SYSTEM

- Participate in time-bound coding competitions
- Create private or public contests with custom settings
- Real-time leaderboards for competitive ranking
- Track contest history and user ratings
- Support for multiple contest formats and durations

CODE EDITOR FEATURES

- Code execution with Multiple Language Support
- Real-time error and output feedback
- Auto-save to prevent code loss
- Output display with execution time and status
- Multi-language support for diverse coding preferences

DISCUSSION FORUM

- Create and join discussions on coding topics
- Share solutions, ideas, and different approaches
- Ask questions related to specific problems
- Comment on existing threads for collaborative learning
- Foster community engagement and knowledge sharing

USER PROFILES

- Track individual problem-solving progress
- View detailed submission history and results
- Monitor contest participation and performance ratings
- Receive personalized problem recommendations
- Access user stats, badges, and achievements

SECURITY FEATURES

- Secure user authentication using JWT
- Password hashing implemented with bcrypt
- Protected API endpoints to prevent unauthorized access
- Input validation and sanitization to prevent injections
- Measures to ensure contest integrity and fairness

RESPONSIVE DESIGN

- Optimized for seamless use on desktop, tablet, and mobile
- Adaptive layouts ensure readability on all screen sizes
- Consistent and intuitive user experience across devices
- Touch-friendly interface elements for mobile usability

FUTURE ENHANCEMENTS

- Integration with popular coding interview platforms
- AI-powered code review, feedback, and smart suggestions
- Support for team-based coding contests and collaboration
- Advanced analytics to track and boost learning progress
- Addition of more programming languages and frameworks
- Interactive tutorials and structured learning paths

CONCLUSION

- A comprehensive platform for coding practice and learning
- Engaging features like discussions, contests, and progress tracking
- Intuitive interface backed by a modern tech stack
- Secure, scalable, and built for future expansion